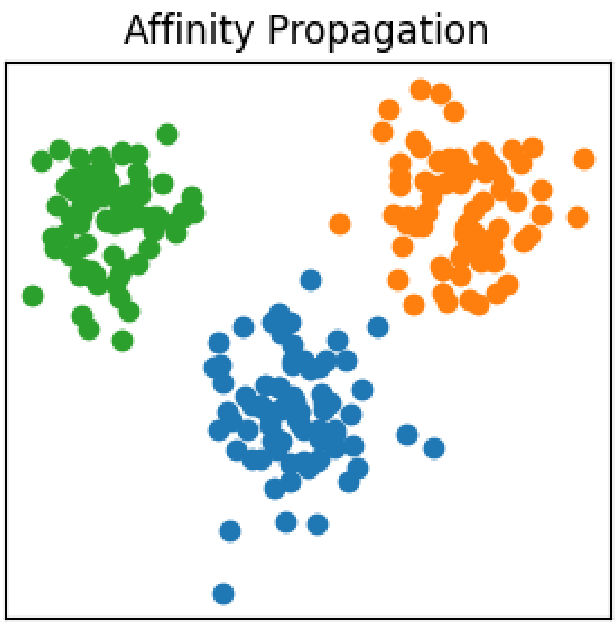
Machine Learning Clustering Handbook

Affinity Propagation



What does it do? Finds important example points and builds clusters around them.

Real-life Example: Grouping students around the best team leader.

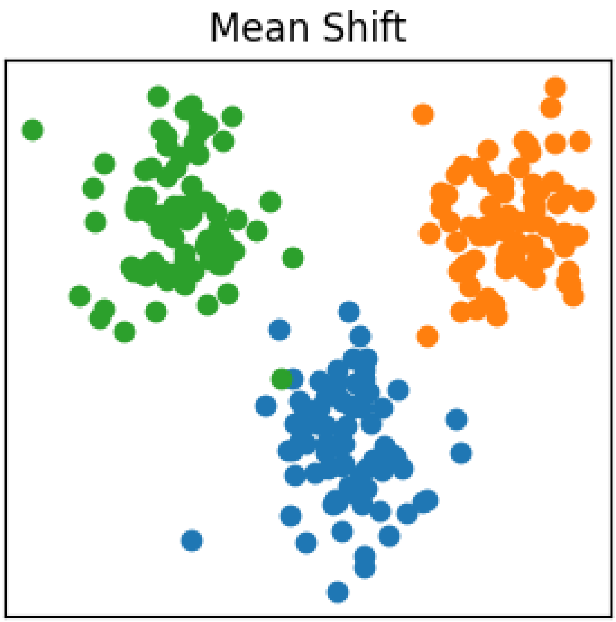
Python Code:

```
from sklearn.cluster import AffinityPropagation

model = AffinityPropagation()
labels = model.fit_predict(X)
```

Advantages	Disadvantages
No need to choose cluster count	Slow for large datasets
Accurate clustering	Uses more memory

Mean Shift



What does it do? Moves points toward crowded areas to form groups.

Real-life Example: People walking toward the busiest shop in a market.

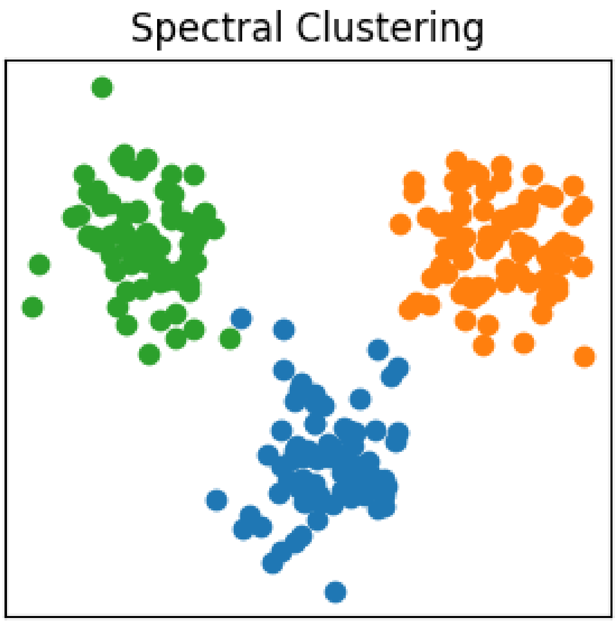
Python Code:

```
from sklearn.cluster import MeanShift

model = MeanShift()
labels = model.fit_predict(X)
```

Advantages	Disadvantages
Finds clusters automatically	Slow on large data
Works for odd shapes	Difficult in high dimensions

Spectral Clustering



What does it do? Uses graph connections to separate related data.

Real-life Example: Finding friend groups in social media.

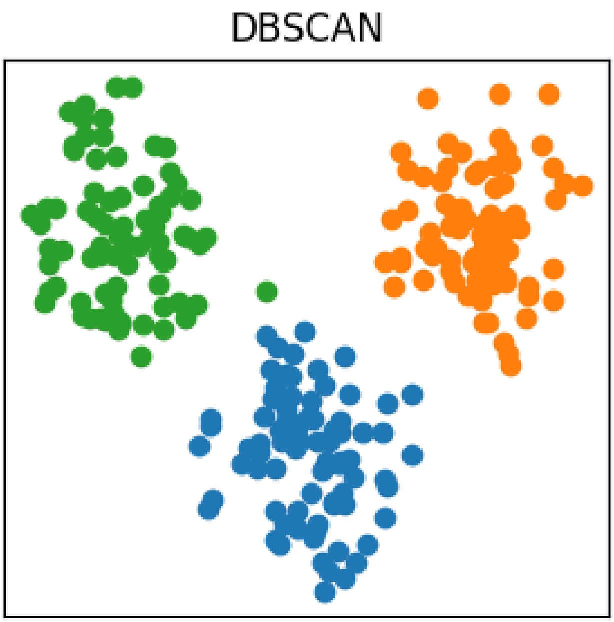
Python Code:

```
from sklearn.cluster import SpectralClustering

model = SpectralClustering(n_clusters=3)
labels = model.fit_predict(X)
```

Advantages	Disadvantages
Good for complex patterns	Needs cluster number
Works with graph data	Computationally expensive

DBSCAN



What does it do? Finds dense groups and removes noisy points.

Real-life Example: Detecting crowded areas in a city map.

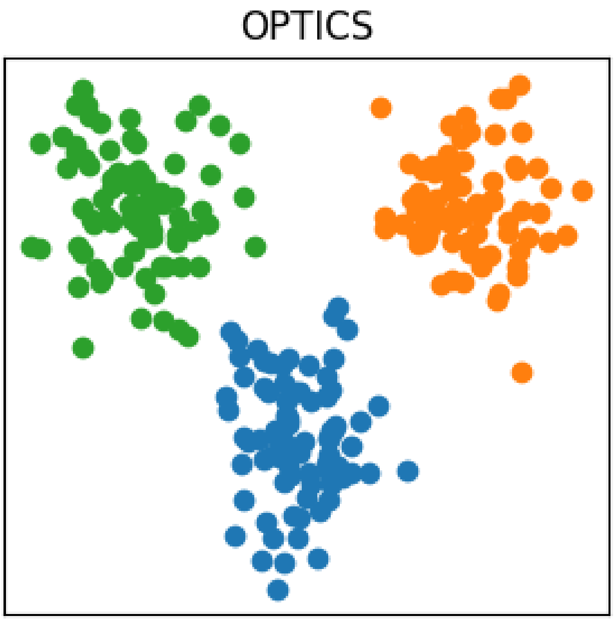
Python Code:

```
from sklearn.cluster import DBSCAN

model = DBSCAN(eps=0.5, min_samples=5)
labels = model.fit_predict(X)
```

Advantages	Disadvantages
Detects noise	Sensitive to settings
Works for irregular shapes	Bad for varying densities

OPTICS



What does it do? An advanced DBSCAN that handles different densities better.

Real-life Example: Finding both villages and cities on a map.

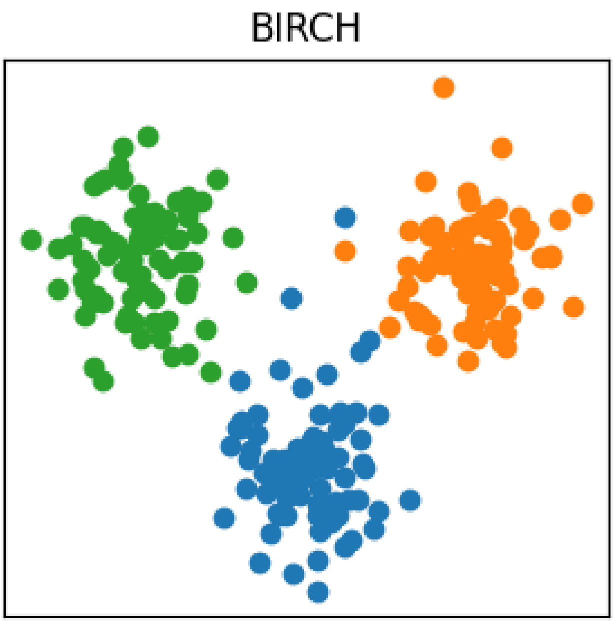
Python Code:

```
from sklearn.cluster import OPTICS

model = OPTICS(min_samples=5)
labels = model.fit_predict(X)
```

Advantages	Disadvantages
Handles varying densities	More complex
Flexible clustering	Slower than DBSCAN

BIRCH



What does it do? Compresses big datasets into summaries before clustering.

Real-life Example: Studying short notes instead of a huge textbook.

Python Code:

```
from sklearn.cluster import Birch

model = Birch(n_clusters=3)
labels = model.fit_predict(X)
```

Advantages	Disadvantages
Fast for big datasets	Works mainly with numeric data
Memory efficient	Less effective for complex shapes

Quick Summary Table

Algorithm	Best Used For
Affinity Propagation	Automatic cluster finding
Mean Shift	Finding dense regions
Spectral Clustering	Complex graph-like data
DBSCAN	Noise detection
OPTICS	Different density clusters
BIRCH	Large datasets